

① Apriori Algorithm - Online shopping Transactions

Pseudocode

INPUT : Transaction Database D

Minimum support threshold minsup

OUTPUT:

Frequent itemsets

BEGIN:

- 1- Generate C1 containing all individual products
 - 2- Count the no. of transactions
 - 3- Remove items whose support is less than min sup
 - 4- Store the remaining items as L1
 - 5- Set k = 2
 - ⑥ while $L(k \neq \emptyset)$ is not empty do
 - ⑦ combine $L_1, L_2, L_3, \dots$ to obtain all freq itemsets
 - 8- Return frequent itemsets
- END

Candidate Generation and Pruning

C1:

(Mobile) $\rightarrow 4/5 = 80\%$

(Corphones) $\rightarrow 3/5 = 60\%$

(Charger) $\rightarrow 3/5 = 60\%$

C2:

(Mobile, Corphones) $\rightarrow 3/5 = 60\%$

(Mobile, Charger) $\rightarrow 2/5 = 40\%$

(Corphones, Charger) $\rightarrow 2/5 = 40\%$

FP Growth Algorithm - Mining Interesting Patterns

Procedure

Algorithm : FP Growth

Input : Transaction Database D

Min support count minsup

Begin

1. Scan the database D
2. Count the frequency of every item
3. Remove whose frequency is less than minsup
4. Sort the remaining items in descending order of freq
5. Create an empty FP-TREE
6. For each transaction T in D DO
7. Create the header table containing each frequent item and links to its tree nodes
8. For each item in the header table DO
9. Combine all generated paths
10. Return freq items

FP TREE CONSTRUCTION

- + All items satisfy the minimum support count of 2
- + Order items by frequency

Pg: 4-12 - Data Science

③

Association Rule Generation - Purchased services

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} \times 100$$

Algorithm

1. For each frequent itemset I in F DO

For each set x of I DO

$$Y = I - x$$

calculate : confidence = $\text{Support}(I) / \text{Support}(x)$

If confidence $\geq$ min conf then

Generate rule $x \rightarrow Y$

Add rule to output

2. Return all strong association rules

END

Rule Generation

	<u>Confidence</u>	<u>Result</u>
Internet streaming	75%	Accepted
streaming $\rightarrow$ Internet	75%	Accepted
streaming $\rightarrow$ cloud storage	50%	Rejected
cloud storage $\rightarrow$ streaming	66.67%	Rejected

4. Constraint - Based Frequent Itemset Mining

Assume min support = 40%.

Min support count = $40\% \times 5 = 2$

Pseudocode :

INPUT : Transaction database D

Min support minsup

Req item = Python

OUTPUT : Frequent itemsets containing Python

Begin

1. Scan D and identify all items
 2. Generate candidate 1-itemset
 3. Remove every itemset that does not contain Python
 4. Count support of remaining candidates
 5. Remove candidate whose support is less than minsup
 6. Set $k=2$
 7. while candidate itemsets exist do
 8. Return all frequent itemsets containing Python
- END

Final Frequent Itemsets

* { Python } - 80%

* { Python, Database } - 40%

EFFECT OF CONSTRAINTS

The Python constraint allows the algorithm to avoid generating itemsets that do not contain Python counting operations, memory usage and overall computational complexity

5. Multilevel Association Rule Mining - Hospital Treatment

Pseudocode

Begin

- 1- Define the treatment hierarchy H
- 2- Start mining from the highest level of hierarchy
- 3- Count the support of treatments at current level
- 4- Remove treatment T whose support is less than $minsup$
- 5- Generate freq itemsets at the current hierarchy
- 6- Generate Association Rules from the frequent itemsets
- 7- Calculate confidence for each rule
- 8- Move to the next lower level in hierarchy
- 9- Repeat the steps 3-8 until all required hierarchy levels are processed

Applying Hierarchy

Level 1 - Healthcare

$$\text{Support (Healthcare)} = \frac{5}{5} = 100\%$$

Treatment types

$$\text{Support (cardiology)} = \frac{3}{5} = 60\%$$

$$\text{Support (orthopedics)} = \frac{2}{5} = 40\%$$

$$\text{Support (ECG)} = \frac{2}{5} = 40\%$$

$$\text{Support (cardiology)} = \frac{2}{5} = 40\%$$

$$\text{Confidence} = \frac{\text{Support (cardiology, ECG)}}{\text{Support (cardiology)}} \times 100$$

$$= \frac{\frac{2}{5}}{\frac{3}{5}} \times 100 = 66.67\%$$

* ECG is associated with cardiology at the lower treatment level, while Healthcare provides the broader Parent-level content